

New Spider Mites in Ontario

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<https://bio.site/grbiclab>

Emergent mite pest species found infesting crops in southwestern Ontario along the two-spotted spider mite, showing a shifting pest landscape.

Increasing complexity of mite pest community in Ontario

The two-spotted spider mite (*Tetranychus urticae*) infests over 1,100 hosts, resists all pesticides, and is Ontario's main mite pest. Recently, we found two new mite species alongside *T. urticae* in Ontario crops, indicating a changing pest landscape and urgent need for diversity and pest potential assessment (Fig. 1).

New pest species are widespread across Ontario fields and greenhouses

Field surveys have shown a wide distribution of the species across various agricultural systems in Ontario. Although *T. urticae* is the predominant species, several populations have mixed species (Fig. 2).



T. urticae
two-spotted spider mite



T. turkestanii
four-spotted spider mite
not reported in Ontario previously



Tetranychus sp.
red spider mite
not described previously

Figure 1. Spider mite species present in Ontario.

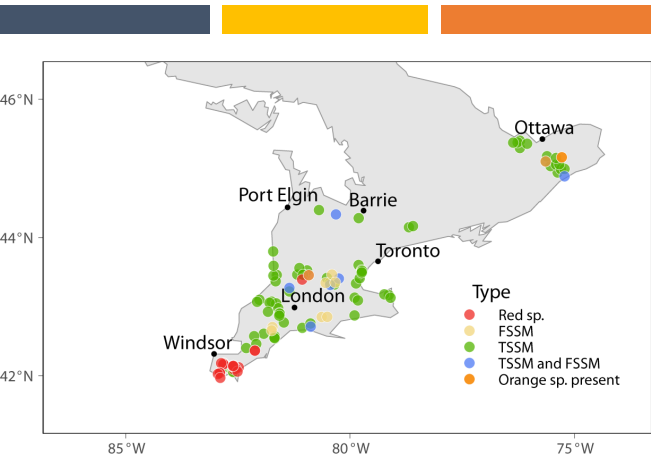


Figure 2. Distribution of emerging mite species across various sites, including field and greenhouse crops in southwestern Ontario.

The new mite species shows distinct morphology. Their morphology enables quick identification: the new *Tetranychus* sp. (red spider mite) is distinguishable by its red color and six spots, while *T. turkestanii* is smaller than *T. urticae* and usually has four spots (Fig. 3).

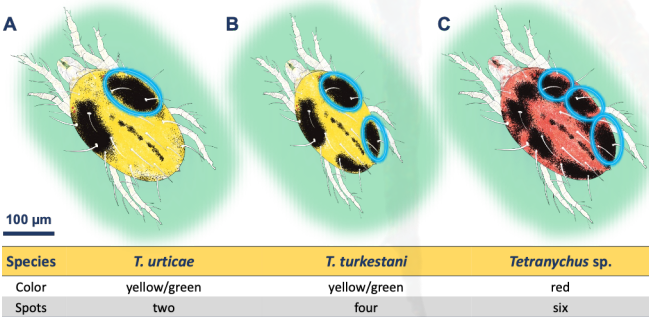


Figure 3. On-site identification of three spider mite species found in Ontario. A. *T. urticae* (two-spotted spider mite, TSSM); B. *T. turkestanii* (four-spotted spider mite, FSSM); C. *Tetranychus* sp. (new red spider mite).

Biological characteristics vary across mite species

Tetranychus sp. has greater longevity, but *T. turkestanii* has a faster developmental time and greater fecundity than *T. urticae* (Fig. 4).

They show the potential to outcompete *T. urticae* and establish dominance, posing an imminent risk of control failure.

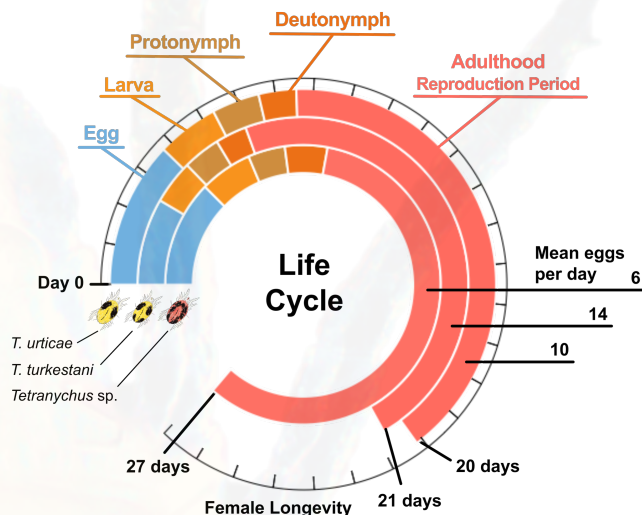


Figure 4. Life cycle, average fecundity, and longevity of different mite species; phenological stages are represented by different colors.

Different mite species induce distinct leaf damage

The pattern of damage varies among species. *Tetranychus* sp. induces greater local damage (chlorosis), whereas *T. turkestanii* induces greater systemic damage (necrosis), culminating in leaf drop (Fig. 5A-B).

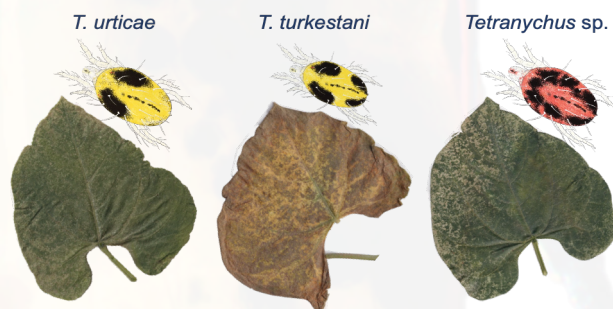


Figure 5A. Damage on bean leaves caused by three species of spider mites. From left to right: *T. urticae*, *T. turkestanii*, and *Tetranychus* sp.

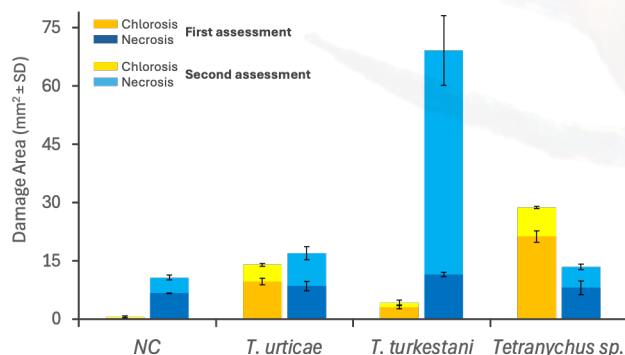


Figure 5B. Leaf damage caused by feeding from different species of mites collected in Ontario. The colors indicate the types of damage (chlorosis vs. necrosis), with dark shades for damage two days after infestation and light shades for damage five days after infestation.

Take Home! Emerging spider mite threats in Ontario

Although *Tetranychus urticae* remains Ontario's dominant spider mite, new species, such as *Tetranychus turkestanii* and a red *Tetranychus* sp., are now widespread across fields and in greenhouses.

These species differ in appearance, reproduction, and the leaf damage, suggesting a shifting pest community that may challenge current management practices.

It is necessary to evaluate the effectiveness of existing and new management strategies against the emerging mite pest complex to tailor management strategies.

Are you having issues with spider mites? Send us samples, and we can help you!

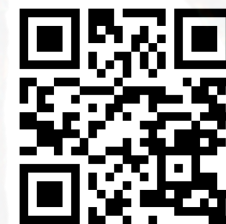
For over a decade, the Grbić Lab at Western University has partnered with growers and stakeholders to study spider mites in Ontario and promote sustainable agriculture.

To submit mite samples:

- Collect infested leaves
- Pack and label them carefully
- Send them to our address
- Notify us once they're shipped

If you see **unusual mites** on your plants, please **send them to Grbić Lab!**

For instructions, please **scan the QR code** →



Acknowledgments



Agriculture and
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PLANTPRODUCTS



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